Assignment-5

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⊙ Title :- Singly Linked List Operations.

⊙ Problem Statement :-

Department of Computer Engineering has student's club named 'Pinnacle Club'. Students of second, third & final year of department can be granted membership on request. Similarly, one may cancel the membership of club. First node is reserved for president of club & last node is reserved for secretary of club. Write C program to maintain club member's information using singly linked list. Store student PRN & name. Write functions to : (a) Add & delete the members as well as president or even secretary. (b) Compute total number of members of club. (c) Display members. (d) sorting of two linked list. (e) Merging of two linked list. (f) Reversing using the pointers.

⊙ Objective :-

1) To understand the concept of a singly linked list & dynamic memory allocation using pointers in C.
2) To perform insertion, deletion, display, count nodes, sort, reverse operations on a singly linked list.
3) To develop a menu-driven application for workshop enrollment management.

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⊙ Pseudocode:-

START
1) Initialize HEAD = NULL.

2) Repeat until user selects EXIT:-
Display menu and accept CHOICE.

3) Perform the selected operation:-
(a) ADD PRESIDENT:-
Check list is empty, input PRN and name, create node & set HEAD to new node.
(b) ADD MEMBER:-
Input PRN and name, check duplicate PRN, create node and insert before Secretary.
(c) ADD SECRETARY:-
Check if Secretary already exists, input details, create node and insert at end.
(d) DELETE STUDENT:-
Search by PRN, update links, free the selected node
(e) DISPLAY:-
Transverse from HEAD to NULL and display PRN, name and role of every student.
(f) COUNT:-
Transverse the list & increment COUNT.
Display total students.
(g) SORT:-
Apply Bubble Sort by comparing adjacent PRNs and swapping student details until sorted.
(h) MERGE:-
Create a second list, check duplicate PRNs, and

insert its nodes before the Secretary.

(i) REVERSE:

Set PREVIOUS = NULL, CURRENT = HEAD.
While CURRENT != NULL:
NEXT = CURRENT.NEXT
CURRENT.NEXT = PREVIOUS
PREVIOUS = CURRENT
CURRENT = NEXT
Set HEAD = PREVIOUS.

(j) DELETE ALL:

Transverse the list, free every node, and set HEAD = NULL.

(k) EXIT:-

Free all nodes and terminate.

4) Repeat menu until EXIT is selected.

STOP

⊙ Time Complexity:-

Operation	Best case
1) Create a node	O(1)
2) Search student	O(1)
3) Add President	O(1)
4) Add Regular member	O(n)
5) Add Secretary	O(1)
6) Display students	O(n)
7) Count students	O(n)
8) Delete student	O(1)

9)	Sort by Bubble Sort	O(n)
10)	Reverse linked list	O(n)
11)	Merge two lists	O(n+m)
12)	Delete all students	O(n)

⊙ Conclusion:-

In this practical, a Student Club enrollment management system was developed using a singly linked data structures. Operations such as insertion, deletion, transversal, counting, sorting & accessing were successfully implemented. This practical enhances the understanding of linked list operations and their applications in real-world record management systems.

⊙ FAQs:-

1) Why is a linked list preferred over an array? What information is stored in each node of a Circular Linked List?

Ans) A linked list allows dynamic memory allocation & easy insertion & deletion.

2) What is a singly linked list?

Ans) A singly linked list is a linear data structure in which each node contains data and a pointer to the next node

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3) What information is stored in each node?
Ans) Each node stores the student's PRN, name, role & a pointer to the next node.

4) What is dynamic memory allocation?
Ans) Dynamic memory allocation is the process of allocating memory during program execution using functions such as malloc() and calloc().

5) Which function is used to release allocated memory?
Ans) The free() function releases dynamically allocated memory.

6) What is the time complexity of inserting a node at the beginning?
Ans) O(1), because only a few pointer updates are required.

7) What is the time complexity of searching for a student?
Ans) O(n) in the worst case, because each node may need to be checked.

8) How is the linked list reversed?
Ans) The list is reversed using three pointers: previous, current and nextNode.

9) Which sorting technique can be used in a linked list?
Ans) Bubble sort can be used.